THEXRA ENTERPRISE SOLUTIONS

SOLUTION PROPOSAL

Client: Malaysia Infrastructure ... **Project:** Malaysia Infrastructure Regulatory Agency Governme...

Date: October 19, 20...

1. Executive Summary

This executive proposal outlines the deployment of Digital Twin — THEXRA Digital Twin (Government Edition) for Malaysia Infrastructure Regulatory Agency to solve critical operational bottlenecks.

2. Client Challenge

The Malaysia Infrastructure Regulatory Agency experiences severe delays and vulnerabilities in public works stewardship due to paper-based assessments and distant visual estimations across geographically isolated arterial bridge assets. This decentralized manual approach creates operational backlogs in safety certification, hinders early identification of structural fatigue and micro-fractures, and impedes data-

3. Project Objectives

Achieve a 50% reduction in safety inspection reporting turnaround while ensuring zero undetected structural fatigue anomalies across critical remote spans for 51-250 Public Works Civil Engineers and Bridge Safety Inspectors within a 4.5-month deployment.

4. Proposed Solution

An enterprise structural digital twin ecosystem integrating parametric 3D models with IoT strain telemetry and spatial scanning feeds, replacing manual paper evaluations with automated anomaly detection, centralized regulatory compliance logs, and predictive asset lifecycle simulations.

5. User Experience

Intuitive spatial interface engineered specifically for Public Works Civil Engineers & Bridge Safety Inspectors (51-250 active users), accelerating user onboarding and minimizing operational learning curves.

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6. Solution Architecture

A sovereign-cloud hosted structural digital twin infrastructure integrating IoT edge strain gauges, drone LiDAR point clouds, and field spatial telemetric models for real-time bridge health monitoring.

7. Scope of Work

Deploy, configure, integrate, test, and deliver an enterprise-grade THEXRA Digital Twin platform tailored for large-scale bridge structural monitoring for Malaysia Infrastructure Regulatory Agency.

8. Deliverables

Enterprise digital twin bridge management portal
Field engineering mobile tablet application
API integrations for sovereign government data repositories
Standard operating procedure and administrative documentation

9. Implementation Plan

Phase 1: Discovery & Architecture
Phase 2: Spatial Customization
Phase 3: Integration & Testing
Phase 4: Final Deployment

10. Timeline

3-6-months

11. Assumptions

1. Key stakeholder availability for bi-weekly milestone sign-offs.
2. Sandbox environment API credentials provided during Phase 1.

12. Optional Add-ons

1. Premium 24/7 SLA Technical Support
2. Advanced Custom Analytics Dashboard
3. On-Site Staff Training Workshops

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13. Next Steps

Schedule a 30-minute technical scoping review for Malaysia Infrastructure Regulatory Agency.

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